

WORK PROCEDURE

PRO-OP-SM-002A

DRILLING OPERATIONS PROCEDURE WITH THE STOPEMASTER

OBJECTIVE

The objective of this procedure is to define safe work methods to perform drilling operations with the STOPEMASTER

APPLICATION

This procedure applies to all employees of MACHINES ROGER INTERNATIONAL.

INTERPRETATION

The personnel having authority for the interpretation of this document is the Operations Manager. Any request for modification, revision or update must be made to this person.

RESPONSABILITIES

The supervisor is responsible for applying and following this procedure. The driller must follow this procedure and all client's procedures. The Driller is responsible for the day-to-day operation of the drill and other on-site equipment required to support the operation. The driller needs the training for all equipment he operates.

- The Driller is responsible for obtaining the maximum possible quality in any given ground condition, while achieving maximum production and maintaining a safe and efficient work environment;
- The Driller is responsible for the general condition of all site equipment and tools;
- The Driller is responsible for obtaining optimum life out of drilling tools and consumables;
- The Driller is responsible for the location of the collar of the hole, ensuring correct alignment with front and back sides, offsets, depth and setting the drill at the correct angle;

DRILLING OPERATIONS PROCEDURE WITH THE STOPEMASTER

- The Driller is responsible for training new Drillers in safe work methods, if requested;
- The Driller is responsible for complying with all environmental regulations as legislated by local Government;
- The Driller is responsible to submit a shift report and to fill in the job's log book.

PROCEDURE

DRILLING DOWN HOLES, INCLUDING BREAKTHROUGH HOLES

1. Check scale -- vibration from drilling can cause loose;
2. When drilling breakthrough holes, always follow breakthrough procedures (Refer to General Practices of the client or follow the MRI procedure: **PRO-SEC-001A Procedure for breakthrough holes**);
3. Install a casing. Follow the MRI procedure: **PRO-OP-SM-004 Procedure to install a casing with a stopemaster**;
4. Gauge bits if you will be using more than one bit with blade bits. Use a caliper on button bits, if necessary. The bit with the longer carbide indicates it is newer, and has a larger outside diameter;
5. Place the rod into the casing and close the centralizer;
6. Holding rod in closed centralizer, lower rockdrill to thread rod onto chuck adapter;
7. Turn on water;
8. Then rotation to screw the rod in the lug steel;
9. When water is seen coming up the casing, slowly allow rod into hole, watching and anticipating jams;
10. As there will be small rocks and cuttings in the hole, it might be necessary to apply a little hammer until you reach the bottom of the previously drilled hole;
11. Drill rod to completion slowly increasing percussion. You do not need full percussion on first rod;

DRILLING OPERATIONS PROCEDURE WITH THE STOPEMASTER

12. To add a rod:

1. stop rotation,
2. stop hammer, and
3. stop water,
4. close centralizer.

13. Reverse feed so weight comes off of rod or coupling;

14. Reverse rotation slowly with some reversed feed until lug steel comes free from rod;

15. Stop rotation, increase reversed feed and allow drifter to climb to top of slide;

16. Add next rod on, thread onto drill string by hand to almost tight, holding rod below coupling;

17. As you slowly add feed, lower drifter so lug steel comes down onto coupling of added rod;

WARNING!!!
STAND AWAY (2.4 METERS) FROM ROTATING RODS
NEVER WEAR LOOSE CLOTHING!

18. Stop feed;

19. Position yourself behind the control panel;

20. Apply rotation so lug steel and rod thread together;

21. Thread chuck adapter onto rod until tight;

22. Release centralizer;

23. Turn water on. Rods are still turning;

WARNING!!!
AT NO TIME SHALL AN OPERATOR HOLD THE ROD WHILE THE LUG
STEEL IS ROTATING!

24. When water is coming up the casing, apply a little feed so there is some pressure on rods;

25. Slowly turn hammer on half and then on full keeping your eye on coupling. Apply feed until there is no longer any vibration at coupling;

Adjustment of the rotational speed and feed will be necessary at this point.
Regularity of rotation is a clue of drilling performance

DRILLING OPERATIONS PROCEDURE WITH THE STOPEMASTER

26. Never allow the rods to bounce. Bouncing rods may cause premature lug wear. Apply a little more feed as the rods get deeper into the hole;
27. You may have to increase rotation or decrease rotation. As you watch the turning coupling, also watch the feed. In time, you will know if the two are turning together;
28. When you come to the end of rod length, repeat steps 10 to 24;

WARNING!!!
NEVER DRILL DRY. THE DUST YOU CREATE IS BREATHED BY YOU AND
YOUR CO-WORKERS.
IT IS HARMFUL FOR YOUR HEALTH!!!!

29. If you are using more than one bit, pull the rods and change bit. DO NOT OVERDRILL BIT;
30. The Driller should always be consciously timing each rod. When penetration starts to slow, change bit immediately. Do not wait until bit is flat! Always drill with a sharp bit;
31. As you come close to breakthrough:
 - a. Flush the hole,
 - b. Allow the water to come back up and
 - c. Continue drilling. Don't allow too much cuttings in the hole at breakthrough; you may jam your rods;
32. At end of shift, always wash off machine and oil up.

KEEP THE WORKPLACE CLEANED

Most poor quality holes are caused from:

- Poor set-up;
- Poor collaring techniques, e.g.: collaring with casing bit;
- Too much feed pressure at beginning of hole: Always drill at least the first 2 rods with minimum feed pressure and minimum hammer;
- Too much feed pressure in structured, faulted ground: Do not use excessive hammer or feed in soft ground; it will not drill any faster;
- Dull bits, never overdrill bits! It causes increased deviation, reduces bit life and slows productivity. It takes less time to drill a hole with three sharp bits than two dull bits.

DRILLING OPERATIONS PROCEDURE WITH THE STOPEMASTER

Loosening rods on non-breakthrough holes:

- Flush hole with air;
- Turn rotation off;
- Water back on;
- Put the Hammer "ON" until water is seen at rod joints and listen for change in percussion sound when joints are loose;
- Blow hole again.

Loosening rods on breakthrough holes:

- Ensure hole is clean of all cuttings;
- With water on, pull rods up to first coupling in centralizer. Close centralizer on bottom section of coupling or joint;
- Using minimum percussion, hammer joints through centralizer until joints are loose. Shut percussion off and reclamp in joints each time until joints are broken;

WARNING!!!: DO NOT HAMMER COUPLINGS OR JOINTS ON TOP OF CLOSED CENTRALIZERS. RISK OF SERIOUS PERSONAL INJURY FROM FLYING METAL CHIPS AND DAMAGE TO EQUIPMENT WILL RESULT!

- If rods are over-tightened due to stuck rods, broken ground, etc., use break-out wrench or break-out plate to loosen rods;
- Sometime, hitting the stuck rods with a sledge hammer will free them.

SAFE OPERATING PRACTICE ON DRILLS EQUIPPED WITH THE MAST MOUNTED FEED BUTTON.

1. Never approach the feed mast with the rotation engaged;
2. Once the rod that you are adding to the string is properly in place, engage the feed button that is attached to the side of the feed mast;

WARNING!!!: KEEP YOUR FINGERS AWAY FROM THE COUPLING JOINT WHILE BRINGING THE SHANK (LUG STEEL) INTO PLACE.

3. Allow the shank to connect to the added rod then disengage the feed button;

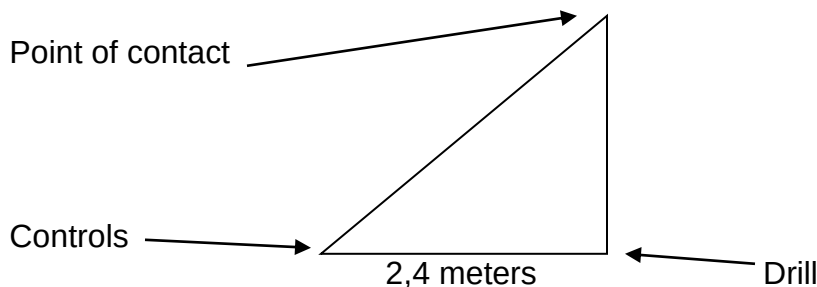
DRILLING OPERATIONS PROCEDURE WITH THE STOPEMASTER

The Feed Button

The feed button is an added feature to the 9000 feed, HX drill. It is an extension of the feed lever on the control panel. With this added feature, there will never be a need to **have the control panel closer than 8 feet (2.4 meters) from the feed mast. All employees must keep 8 feet (2.4 meters) back from operating equipment.**

COLLARING AND DRILLING AN UPHOLE

1. Before moving carrier into set up, add auxiliary lighting possible to assist in visually inspecting ground conditions. Check scale entire back in immediate area;
2. Set up carrier according to the procedure;
3. Choose proper bit and rod length. Make sure length of first rod on uphole is adequate; this will ensure a good solid collar. If back is high, get a longer rod;
4. With the drifter at bottom and centralizer open, use a round rod with a bit and place it on the lug steel;
5. Close the centralizer;
6. Place yourself and control panel at 2,4 meters (8 feet);



7. Bring bit up to roof or backs and apply snug pressure;
8. Turn on the water — enough to eliminate dust;
9. Turn hammer on about 1/4 and apply enough feed so rods don't bounce. Watch for loose and small flying debris;

WATCH FOR THE ROCK SHARDS

WATCH FOR CHANGE IN THE GROUND AND THE GROUND SUPPORT CONDITIONS

10. Once the bit has cut a notch, apply rotation and more feed;

DRILLING OPERATIONS PROCEDURE WITH THE STOPEMASTER

11. Never allow the rods to bounce while collaring, you will break a lug steel or a rod;
12. Once collared:
 - a. **stop hammer,**
 - b. **stop rotation,**
 - c. **stop water,**
 - d. **check scale for loose dislodged from collaring,**
 - e. **Re-check set-up.**
13. Resume drilling until required depth is reached. Use minimum feed pressure for minimum of 3 meters;
14. When adding rods on an up hole, always screw rods into the female end of last rod drilled; or, with rotation in the neutral position, screw rod onto chuck adapter;

WARNING!!!

**NEVER THREAD OR UNTHREAD RODS FROM CHUCK ADAPTER WHILE
ROTATION IS ACTIVATED**

WARNING!!!

**MAINTAIN CONTINUOUS VISUAL SWEEPS OF BACK AND WALLS FOR LOOSE
OR POSSIBLE FALLS OF GROUND**

15. When pulling rods, never allow rods to drop or catch with centralizer. This is dangerous and is equipment abuse. This is a serious infraction and will be dealt with accordingly.

ADDING AND REMOVING RODS

1. After first rod is drilled to complete depth, stop rotation first, then stop hammer. This sequence should break the joint on the coupling;
2. Close centralizer on the coupling, then unscrew the shank from the coupling and bring the drifter to the bottom of the slide;
3. Select next rod from rod rack while drifter is returning to bottom of slide;
4. After the drifter has completely stopped, hand-thread the next rod onto the shank all the way on. Never add a rod to a rotating shank;
5. From behind the control panel, engage the rotation and feed so the next rod goes up to the rod in the centralizer and threads itself into the coupling. (The control panel should be at least 8 feet back from the feed mast);
6. Using the control panel levers, rotate the shank (lug steel) into the added rod and release the centralizer as soon as the threads have completely threaded together;

DRILLING OPERATIONS PROCEDURE WITH THE STOPEMASTER

7. Continue to drill.

TO REMOVE RODS

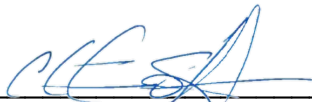
1. After drilling is complete, stop rotation and using low hammer, vibrate the rods to break the joints;
2. Lower the drifter until the second coupling is in the centralizer and close the centralizer on the coupling;
3. If the joints are not broken, break the joints by locking the coupling in the centralizer and applying feed in conjunction with the hammer. Never pound rods on top of the centralizer jaws to break them;
4. Be sure that the centralizer is locked on the coupling, and unscrew the shank (lug steel) from the coupling;
5. Lower the drifter to the bottom of the slide;
6. After the drifter has completely stopped, you can approach the slide to remove the rod. Unthread the rod by hand and return it to the rod rack;
7. From behind the control panel, apply feed so the drifter raise's up the slide to engage the next rod to be removed. With electric hydraulic drills, the control panel should be 8 feet from the feet mast;
8. Thread the lug steel (shank) onto the next rod and release the centralizer. Repeat steps 4 to 8 until string is removed from the hole;

CAUTIONS

When using a helper or trainer, no person is allowed to approach the slide area while any parts are in motion. Trainees must stay near the control panel.

While a helper or trainee is removing or adding a rod at the drill, the trainer must completely remove his hands from the control panel.

At no time shall the drill carrier be used as a work platform



Christian St-Amour
Directeur des opérations